

Screenshots and Brief Summary of Week 4

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks:

Task 1:

1. **Explore Azure Portal** – Azure Portal (portal.azure.com) is a web-based, unified interface provided by Microsoft to create, manage, and monitor all Azure cloud resources without managing real physical server.

Its main components are:

- i) **Dashboard** provides a customizable overview of all your resources and activities upon login.
 - ii) **Search Bar** at the top allows quick navigation to any Azure service like Storage Account, Data Factory, Virtual Machines etc.
 - iii) **Left Sidebar** contains favorites and frequently used services such as Home, Resource Groups, All Resources for quick access.
 - iv) **Resource Creation** is done via the "+ **Create a Resource**" button where 100+ Azure services are available in the Marketplace.
 - v) **Resource Groups** help organize related resources together — we created **RG-group** to hold our Storage Account and Data Factory.
 - vi) **Access Control (IAM)** is available on every resource to manage roles like Reader, Contributor, and Owner for secure access management.
 - vii) **Monitoring Section** provides real-time metrics, activity logs, and alerts for all deployed resources.
 - viii) **Notifications (Bell icon)** at the top shows deployment status — success or failure of any resource creation.
2. **Create a Resource Group** - A Resource Group acts like a folder in Azure that keeps all related resources organized, making it easier to manage, monitor, and delete them together.

Steps for creating Resource Group-

- **Navigation:** Azure Portal → "Resource Groups" → "+ Create"
- **Subscription:** Selected Subscription to manage billing and costs.
- **Name:** Given a unique name to identify the container.
- **Region:** Select the location for resource deployment.
- **Review + Create:** Validated all details and clicked Create to deploy the Resource Group.

Output-

1. **Creating Resource Group-**

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

gaurikapareek1
DEFAULT DIRECTORY

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Home > Resource Manager | Resource groups

Create a resource group ...

BasicsTagsReview + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * Azure subscription 1

Resource group name * RG-group

Region * (Asia Pacific) Central India

PreviousNextReview + create

2. Review + Final Creation-

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

gaurikapareek1
DEFAULT DIRECT

Create a resource

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Microsoft Entra ID

Monitor

Home > Resource Manager | Resource groups

Create a resource group ...

BasicsTagsReview + create

Automation Link

Basics

Subscription Azure subscription 1

Resource group name RG-group

Region Central India

Tags

None

PreviousNextCreate

3. Notification of succesfully creating Resources-

Notifications

More events in the activity log →

Dismiss all

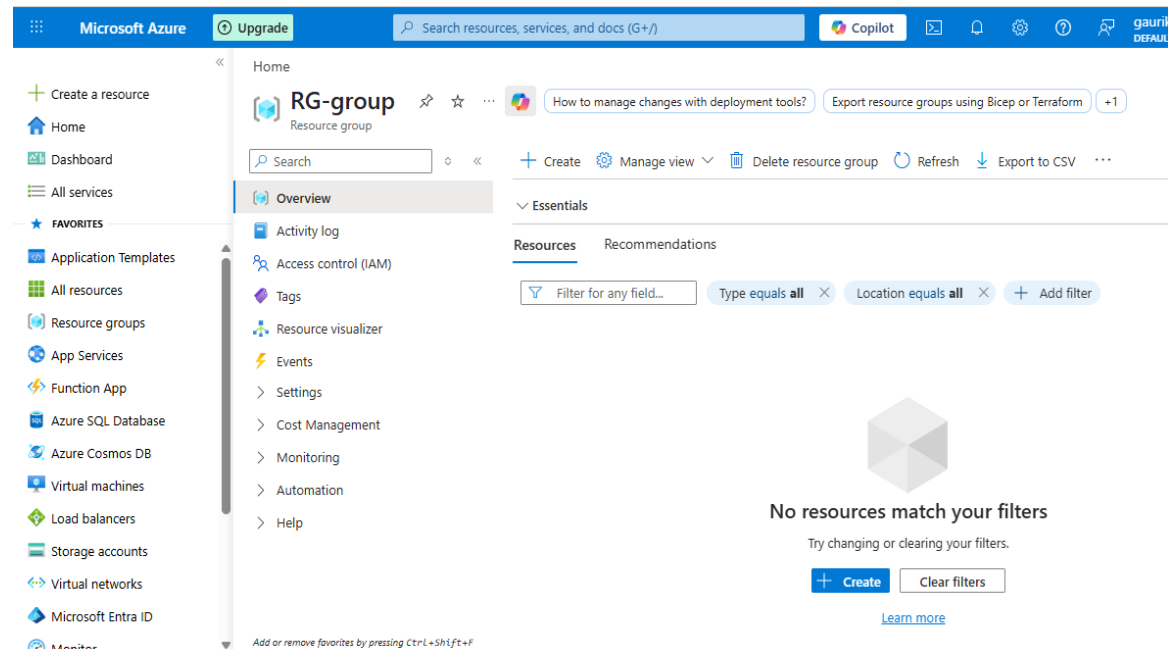
✓ Resource group created

Creating resource group 'RG-group' in subscription 'Azure subscription 1' succeeded.

Go to resource groupPin to dashboard

a few seconds ago

4. **Overview of Created Resource Group-** This overview page shows all resources inside the group along with options like Create, Manage View, Delete Resource Group, Export to CSV and Activity Log.



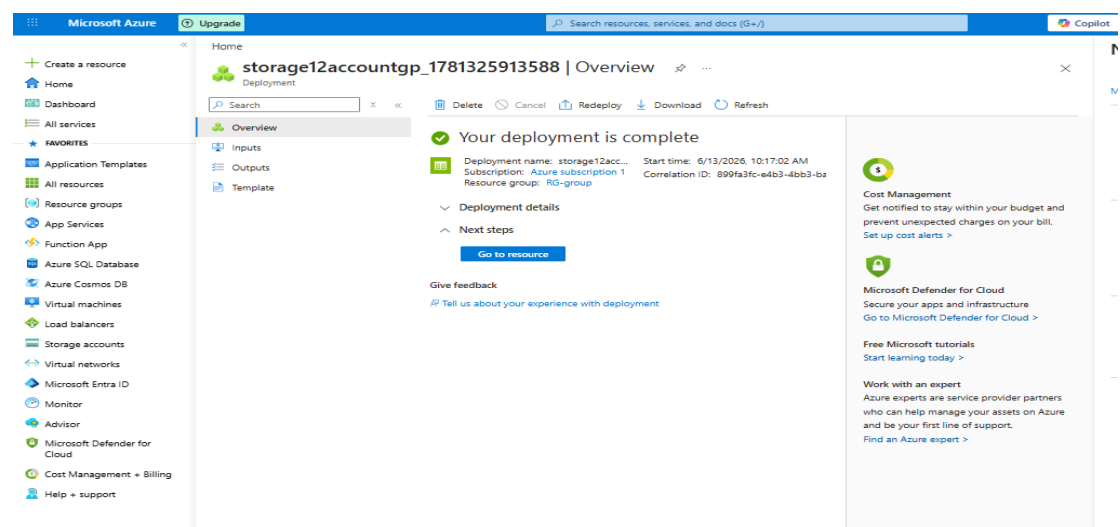
Task 2: Storage Setup

1. **Create a Storage Account-** In this task, we successfully created a **Storage Account** named **"storage12accountgp"** in Azure as shown by the deployment confirmation screen.

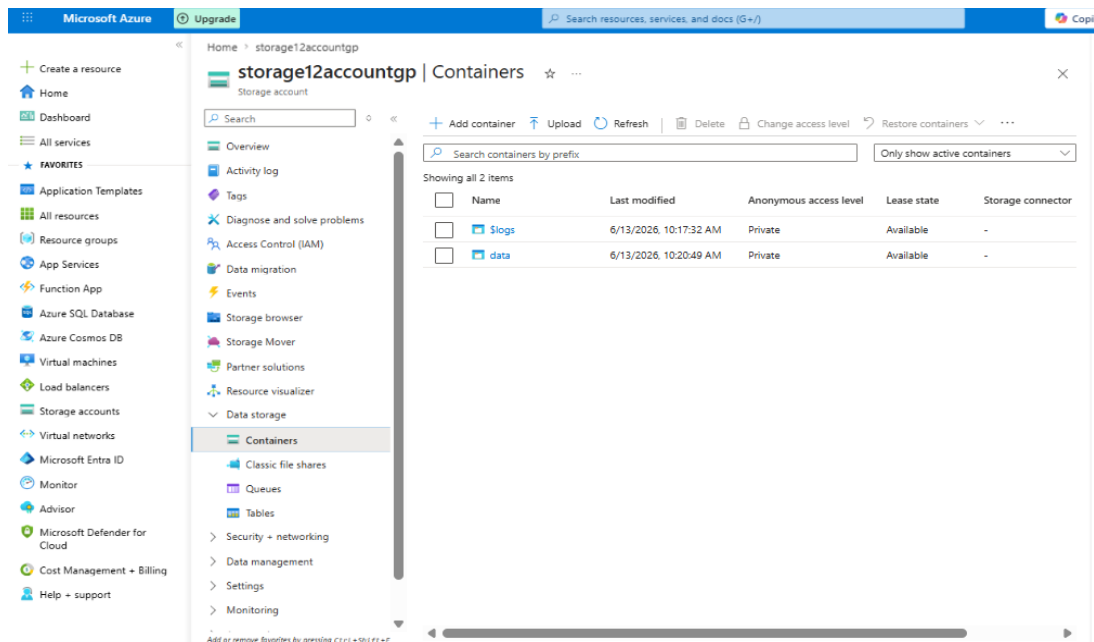
Steps-

- Azure Portal → Storage Accounts → + Create → filled in required details → Review + Create → Create
- In the output shown, “Your deployment is complete” confirmed that Storage Account was successfully provisioned.

Output-

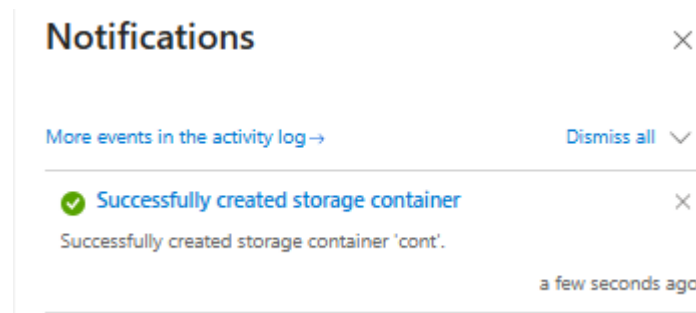


2. **Create a Blob Container** - Now we will create a storage container. Tap on the above “Go to resource Tab”. This looks like –

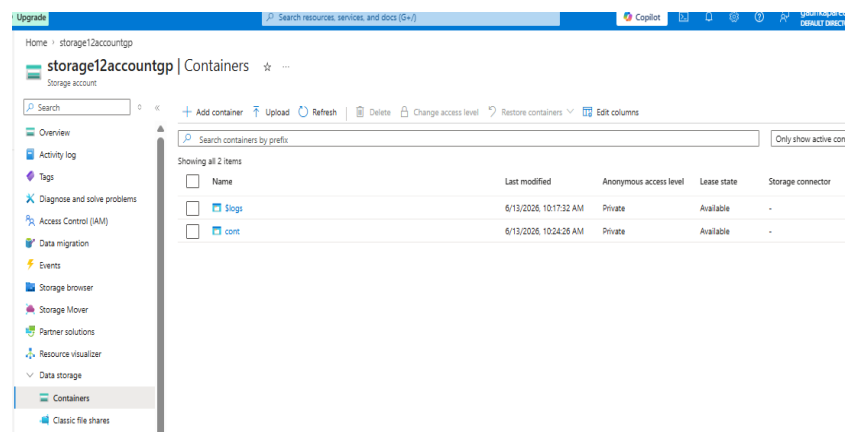


Further steps-

- Data Storage → Containers → New + → Write name of any container you want to create. After creation there is a final notification for creation.

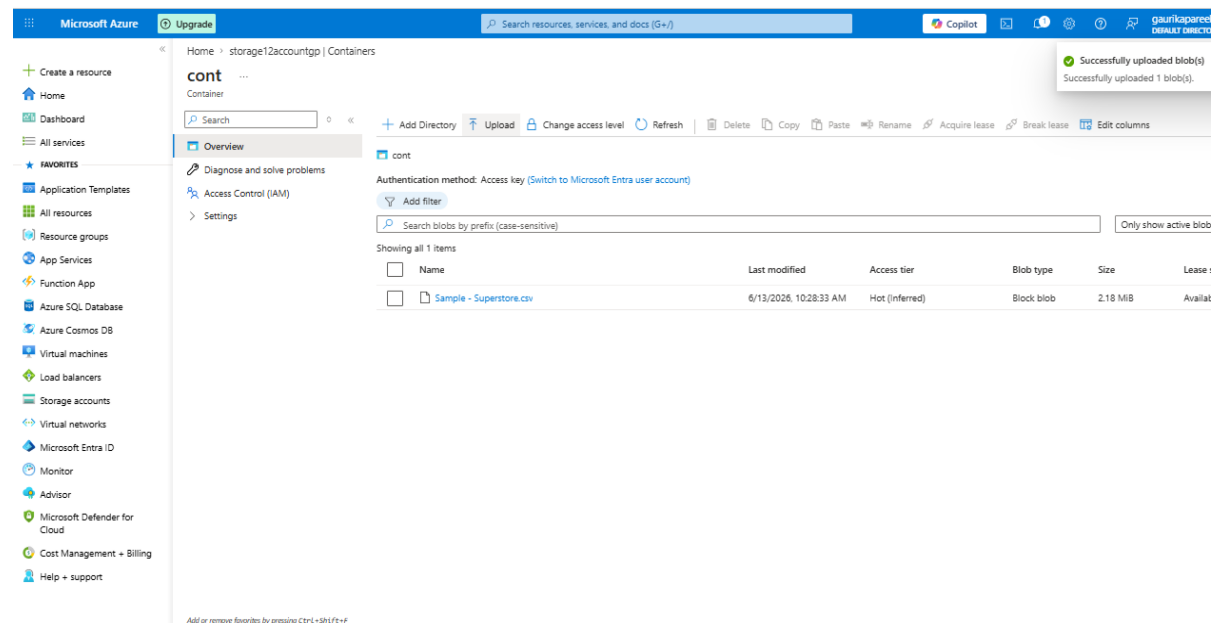


- A container named ‘cont’ is created successfully and will appear here-



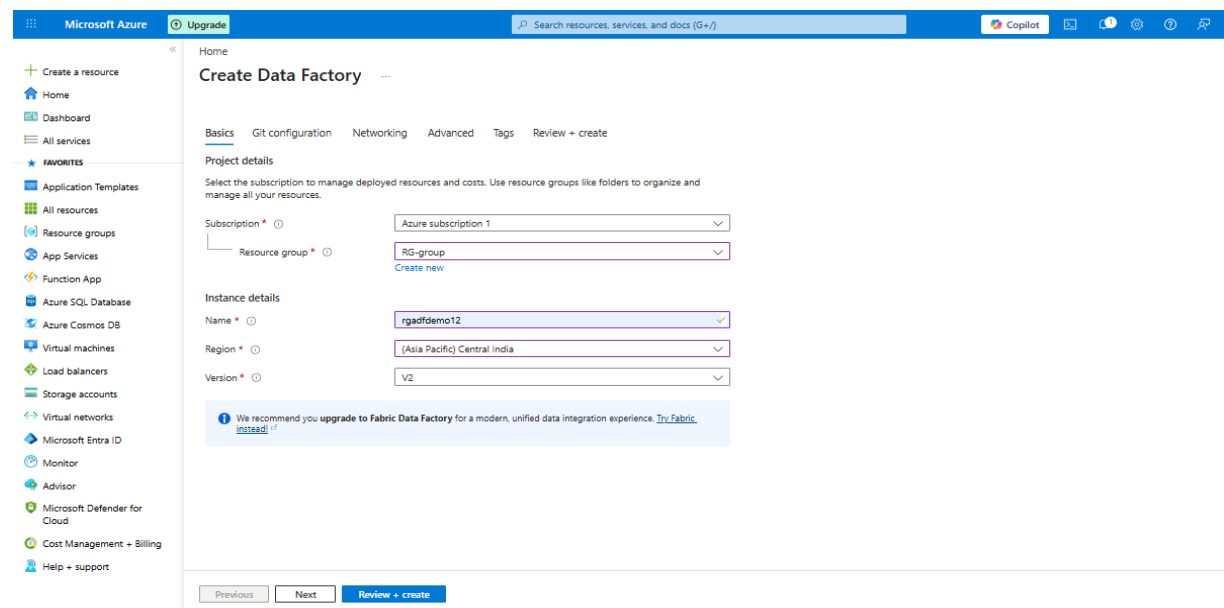
3. **Upload a CSV file-** In the container 'cont', we uploaded a CSV File From our respective device by upload option.

Output- A csv file is successfully uploaded named as "Sample-Superstore.csv".

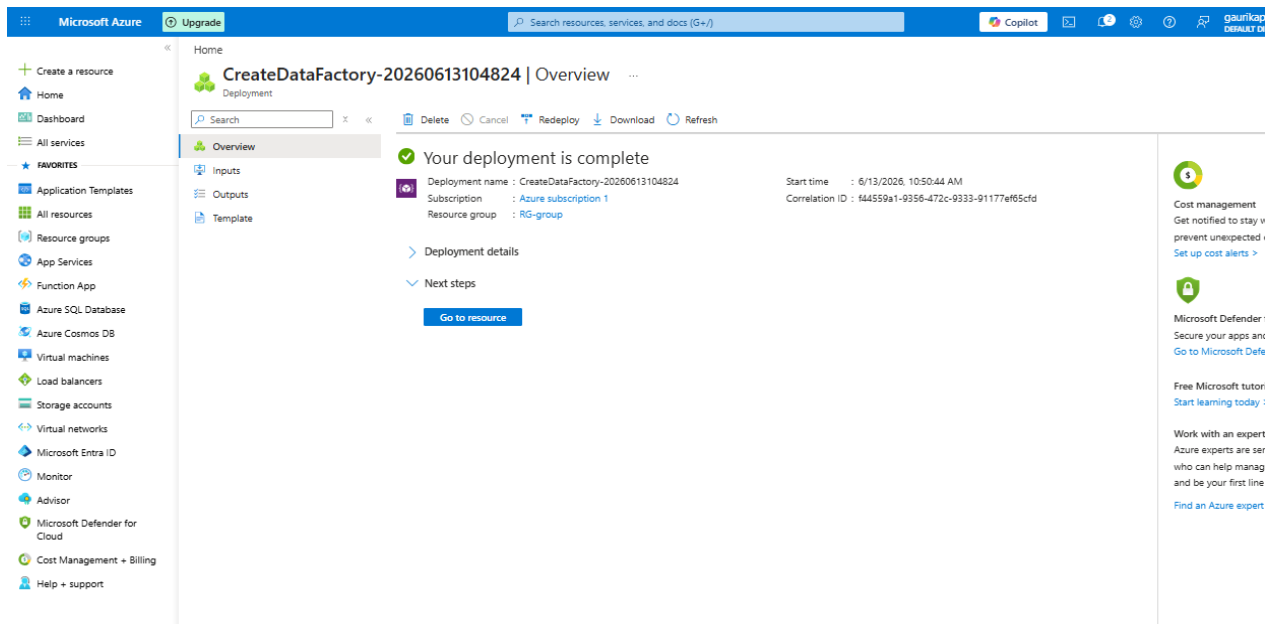


Task 3: ADF Basics –

1. **Create Azure Data Factory** - Go to Azure Portal → + Create a Resource → Search "Data Factory" → Create



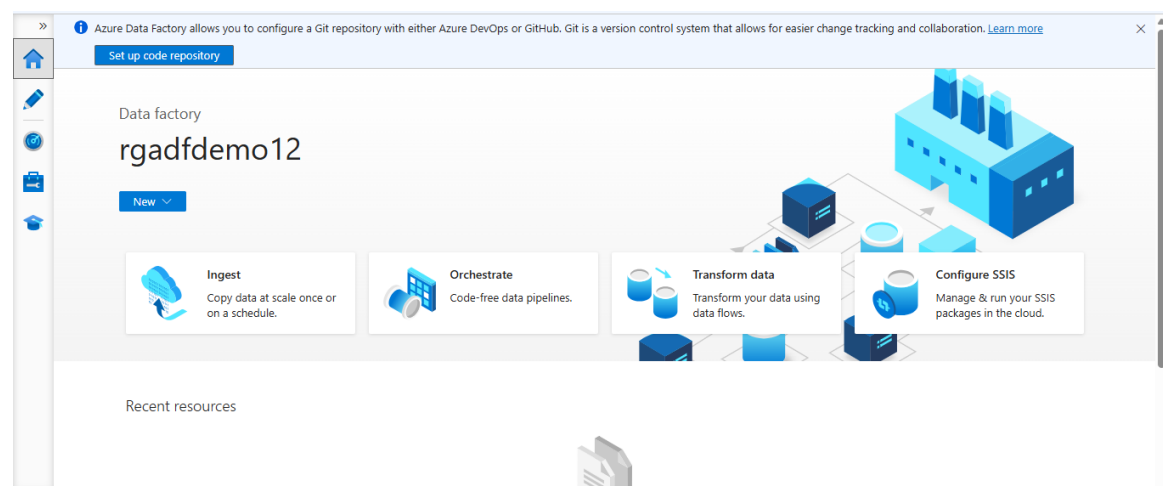
Output – Finally Data Factory is created.



2. Explore ADF UI (Author, Monitor, Manage) - After successfully creating Azure Data Factory, we launched ADF Studio and explored its three main sections:

- **Author (Pencil Icon):** This is the main workspace where we create and manage Pipelines, Datasets, Linked Services, and Data Flows.
- **Monitor (Graph Icon):** This section displays real-time status of all Pipeline Runs, Activity Runs, and Trigger Runs with their execution status like Succeeded, Failed, or in Progress along with duration and run details.
- **Manage (Wrench Icon):** This section is used to manage Linked Services, Integration Runtimes, Triggers, and Git Configuration.

After launching ADF Studio this is the view that contains the above three icons.



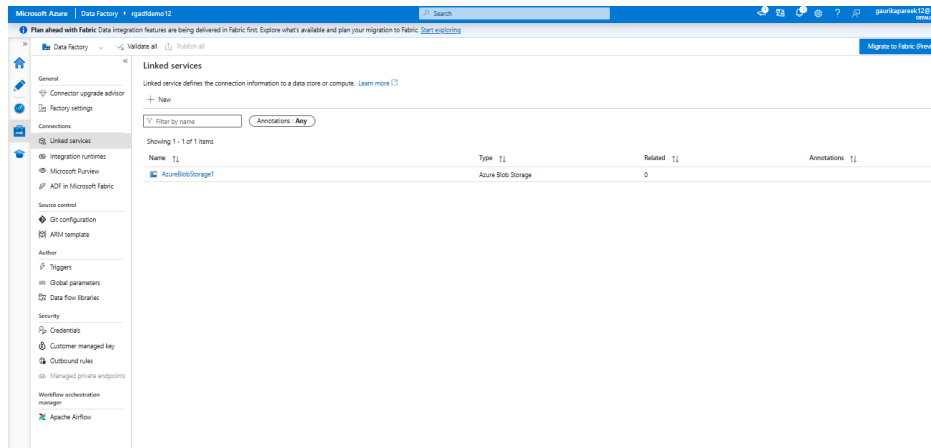
3. Create Linked Service (Blob Storage) -

Steps-

- ☐ **Step 1:** ADF Studio → Manage (Wrench Icon) → Linked Services → + New
- ☐ **Step 2:** Searched "Blob" → Selected Azure Blob Storage → clicked Continue

- ❑ Step 3: Give any name and selected Account Key as authentication method
- ❑ Step 4: Select your respective storage account as the Storage Account
- ❑ Step 5: Clicked "Test Connection" → got "Connection Successful"
- ❑ Step 6: Clicked Create → Linked Service successfully created
- ❑ Step 7: Clicked Publish All → Publish to save changes

Output - Linked Service is successfully created.

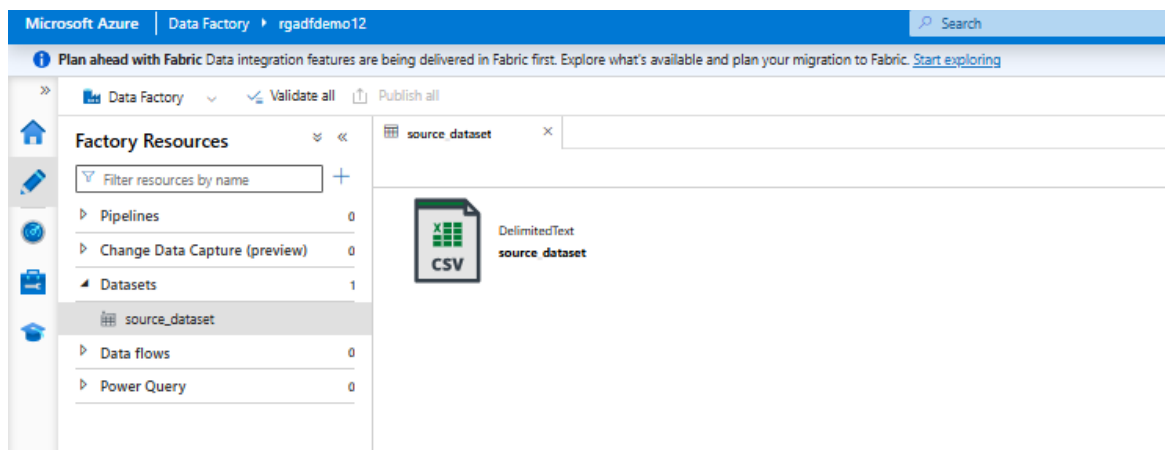


4. **Create datasets (source + destination)** –

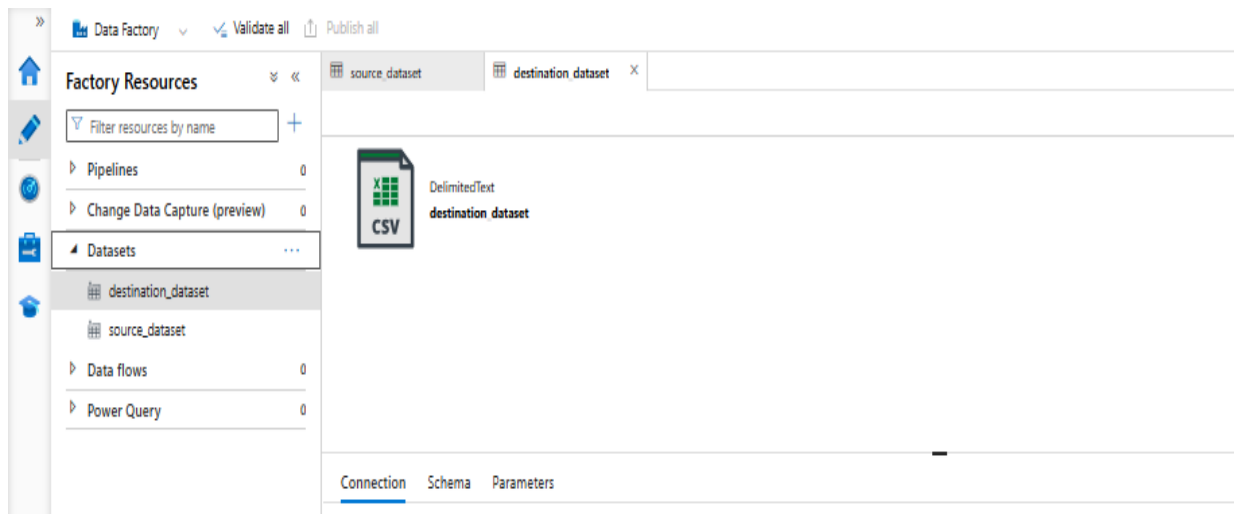
Steps-

- **Step 1:** ADF Studio → Author (Pencil Icon) → Datasets → + New Dataset
- **Step 2:** Selected Azure Blob Storage → delimited text(CSV) → Continue
- **Step 3:** Fill details like Name, linked service, file path, and check the box “First row header” and then Create Source and Destination Datasets individually by Clicking **Publish All** → **Publish** (Repeat the above steps individually for both source and destination).

Output-Source dataset is successfully created.



Output- Destination dataset is successfully created.

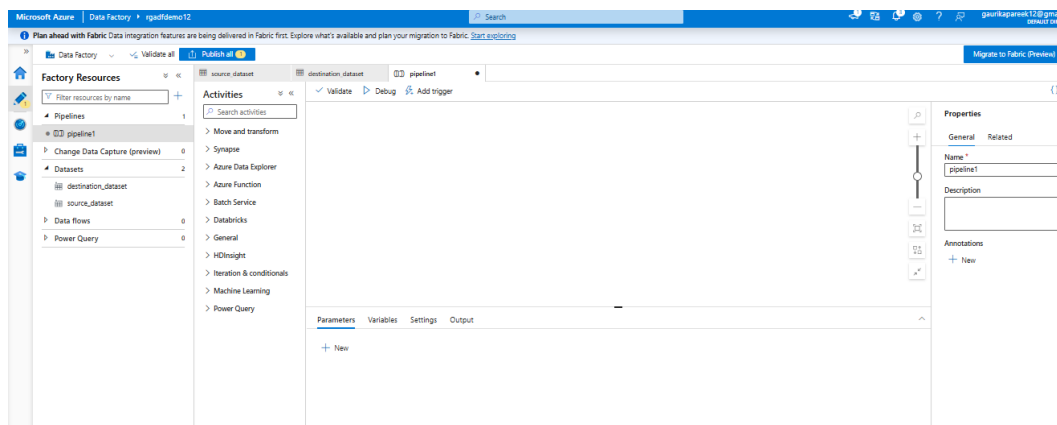


5. **Use Get Metadata activity-**

Steps-

- ☐ **Step 1:** ADF Studio → Author (Pencil Icon) → Pipelines → + New Pipeline
- ☐ **Step 2:** Name the pipeline by any name you want.
- ☐ **Step 3:** In Activities panel → General → dragged "Get Metadata" onto canvas
- ☐ **Step 4:** Clicked on Get Metadata → Settings tab → configure further settings according to the source dataset you had uploaded like item name, type, size.
- ☐ **Step 5:** Click debug to test further once it gets succeeded Click Publish All → Publish.

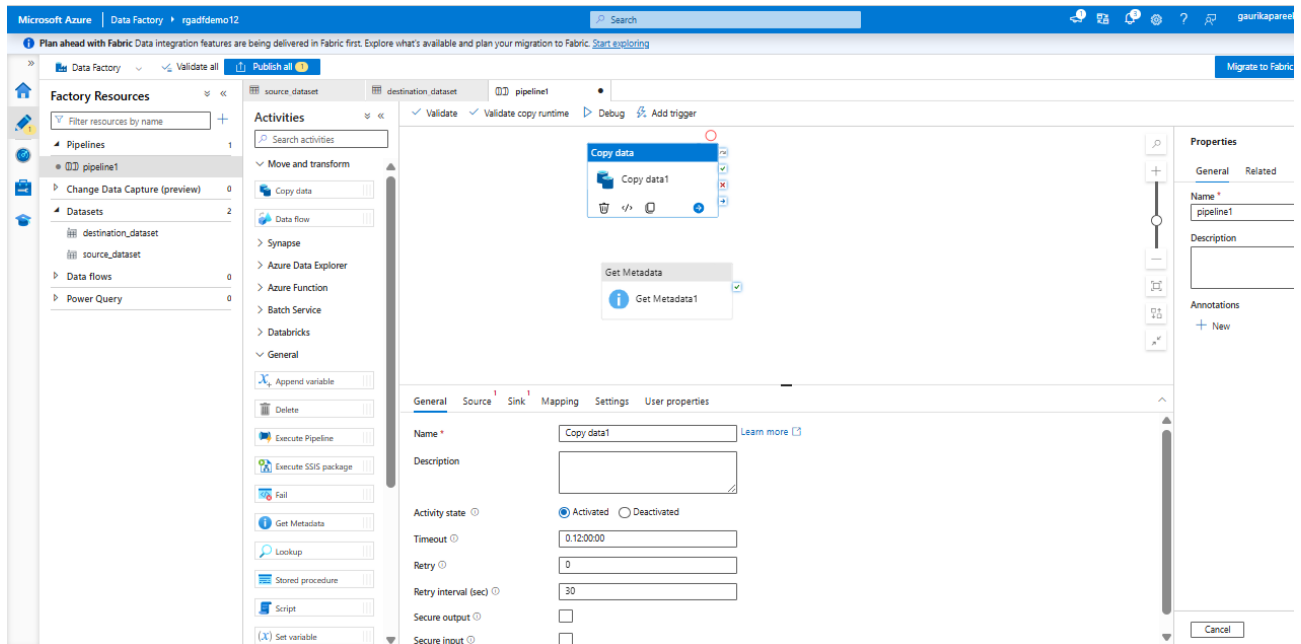
Output- Firstly pipeline is created.



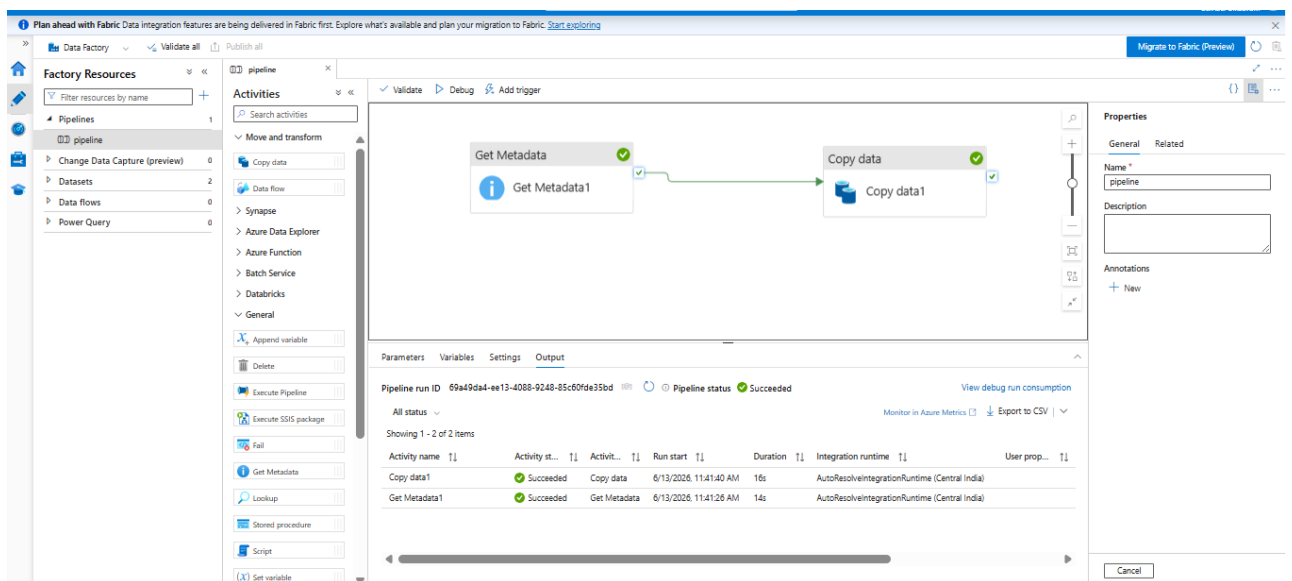
Output- Get Data in canvas.

- ❑ **Step 3:** Activities panel → Move and Transform → dragged "Copy Data" onto canvas
- ❑ **Step 3:** Click Copy data1 and then set Source and Sink as Source and destination Dataset we had created earlier.
- ❑ **Step 4:** Connect Get Metadata1 → Copy Data1 using green arrow.

Output- Copy Data in canvas and set Source and Sink as Source and destination csv we had created earlier.

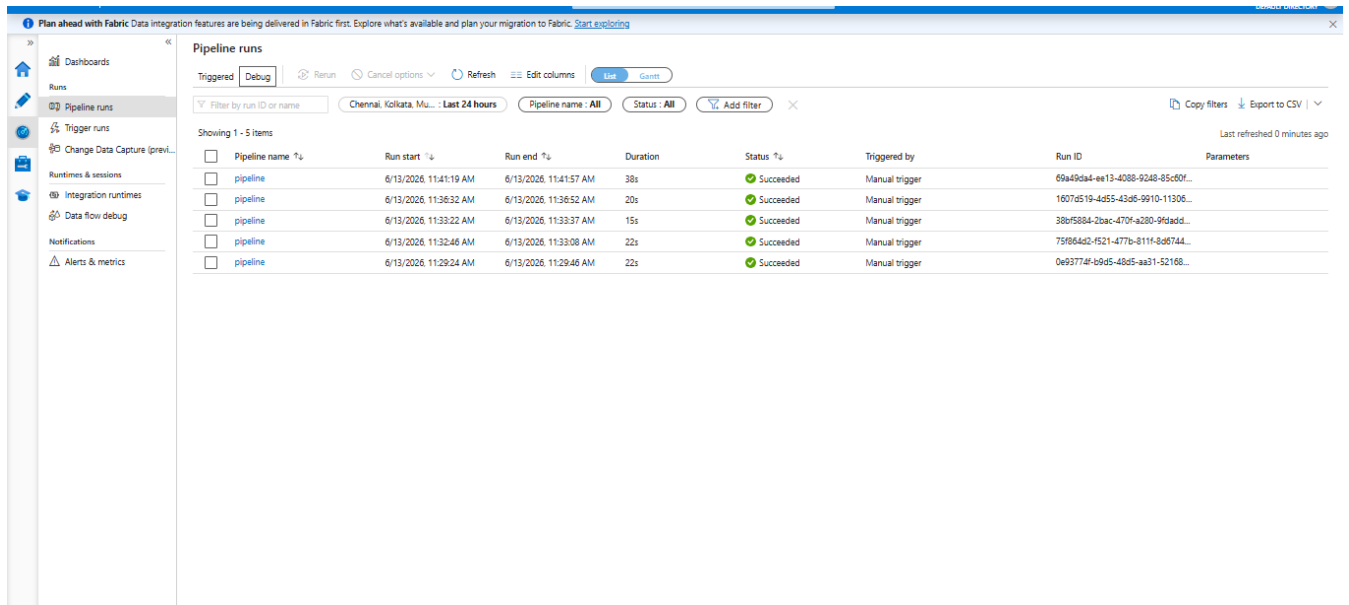


Output- Canvas showing Get Metadata1 → Copy Data1 connected and both activities Succeeded



Task 5: Pipeline Execution

Output- Pipeline "pipeline" was successfully executed multiple times using Debug/Manual Trigger and monitored in ADF Monitor section — all these runs showed Succeeded status with data copied from source to destination.



The screenshot shows the 'Pipeline runs' section in the Azure Data Factory Monitor. The left sidebar contains navigation options: Dashboards, Runs, Pipeline runs, Trigger runs, Change Data Capture (previ..., Runtimes & sessions, Integration runtimes, Data flow debug, Notifications, and Alerts & metrics. The main area displays a table of pipeline runs for a pipeline named 'pipeline'. The table has columns for Pipeline name, Run start, Run end, Duration, Status, Triggered by, Run ID, and Parameters. All five runs shown are in a 'Succeeded' status and were triggered manually. The runs occurred on 6/13/2026 at various times, with durations ranging from 15s to 38s.

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Parameters
pipeline	6/13/2026, 11:41:19 AM	6/13/2026, 11:41:57 AM	38s	Succeeded	Manual trigger	69a49da4-ee13-4088-9248-85c50f...	
pipeline	6/13/2026, 11:36:32 AM	6/13/2026, 11:36:52 AM	20s	Succeeded	Manual trigger	1607d519-4d55-43d6-9910-11306...	
pipeline	6/13/2026, 11:33:22 AM	6/13/2026, 11:33:37 AM	15s	Succeeded	Manual trigger	38bf5884-2bac-470f-a280-96dadd...	
pipeline	6/13/2026, 11:32:46 AM	6/13/2026, 11:33:08 AM	22s	Succeeded	Manual trigger	75f864d2-f521-477b-811f-8d6744...	
pipeline	6/13/2026, 11:29:24 AM	6/13/2026, 11:29:46 AM	22s	Succeeded	Manual trigger	0e93774f-bb9d5-48d5-aa31-52168...	

Task 6: IAM Roles

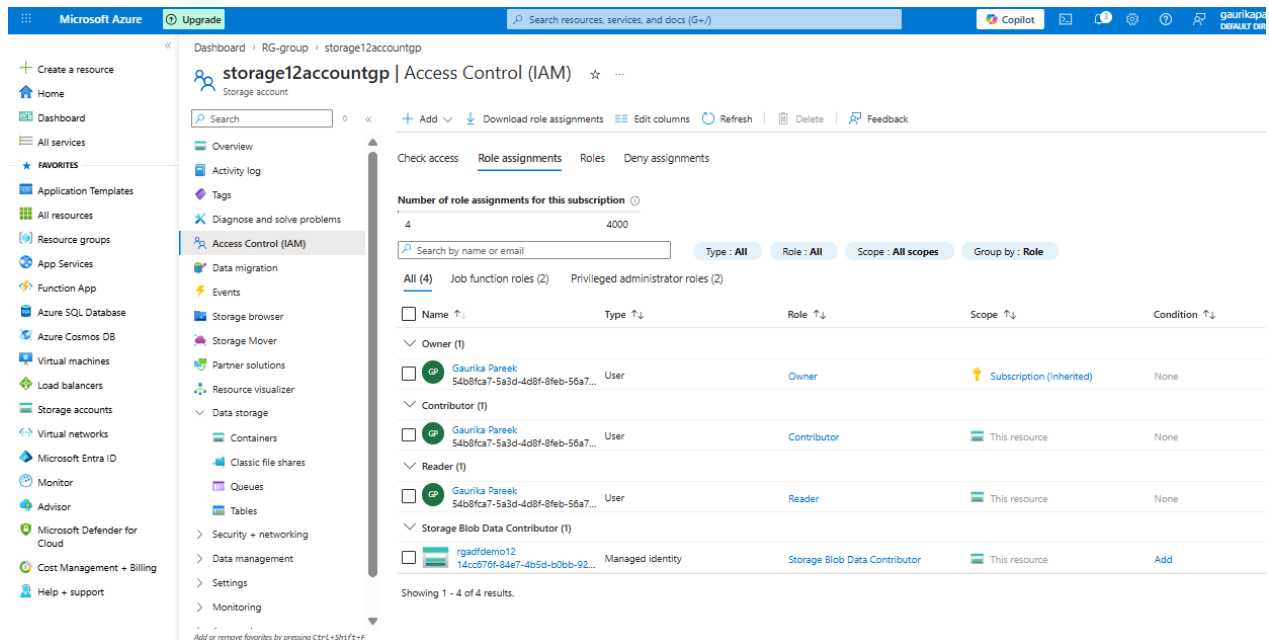
Steps:

- **Step 1:** Azure Portal → Storage Accounts → Your account
- **Step 2:** Left sidebar → Access Control (IAM) → Role Assignments tab
- **Step 3:** Clicked + Add → Add Role Assignment
 - i) Assign roles:
 - Reader
 - Contributor
- **Step 4:** Selected "Reader" → clicked Next
- **Step 5:** Members → + Select Members → select your mail you want to add → Next → Review + Assign
- **Step 6:** Again + Add → Add Role Assignment → Privileged Administrator Roles tab.
- **Step 7:** Selected "Contributor" → clicked Next
- **Step 8:** Members → + Select Members → select your mail you want to add → Review + Assign

Provide ADF Access to Storage:

- **Step 9:** Again + Add → Add Role Assignment
- **Step 10:** Selected "Storage Blob Data Contributor" → clicked Next
- **Step 11:** Members → Managed Identity → select your ADF → Review + Assign

Output- All required IAM roles were successfully assigned on Storage Account— Reader and Contributor roles were given to user and Storage Blob Data Contributor role was assigned to ADF as Managed Identity to ensure secure access between Azure Data Factory and Blob Storage.



Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

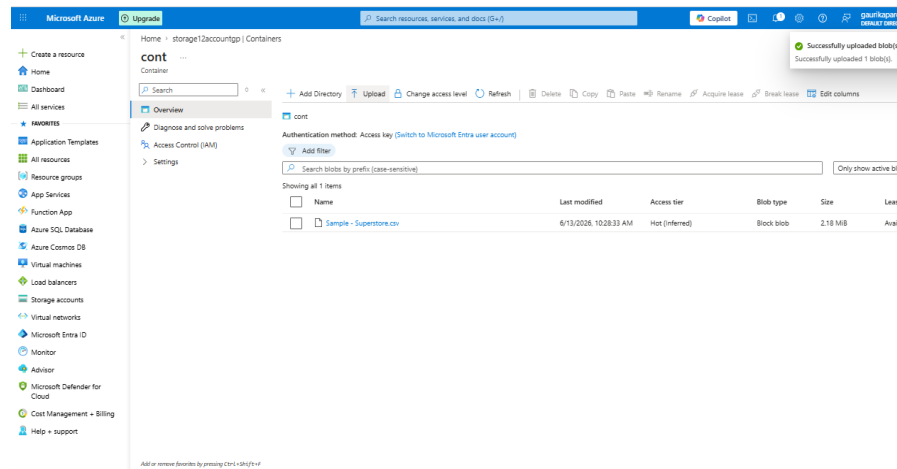
- Pipeline executed successfully
- Data copied to destination
- Metadata validated

Solution for this Project

Steps-

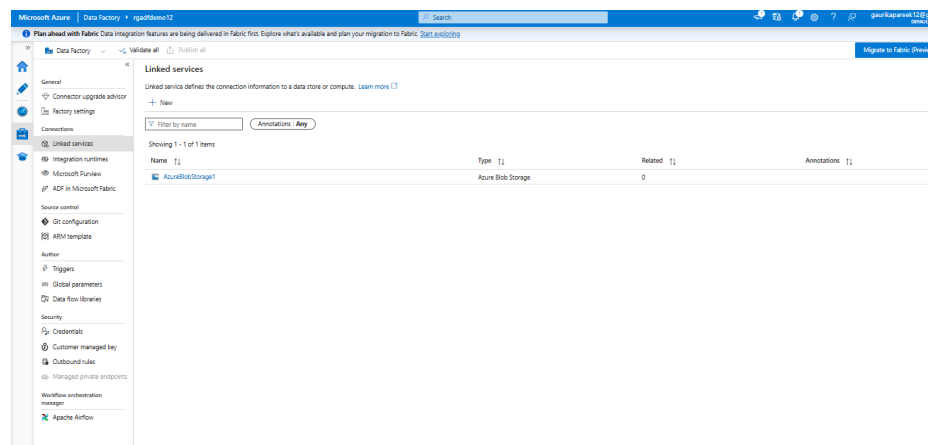
1. Successfully uploaded Sample-Superstore.csv in Blob Container "cont" of Storage Account as the source data file.

Output-



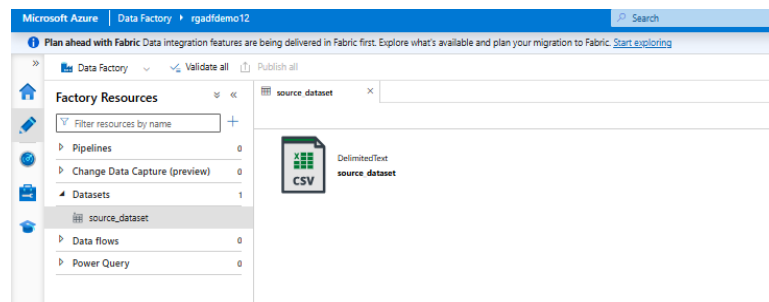
2. Created Linked Service in ADF Studio under Manage → Linked Services to establish connection between ADF and Blob Storage.

Output-



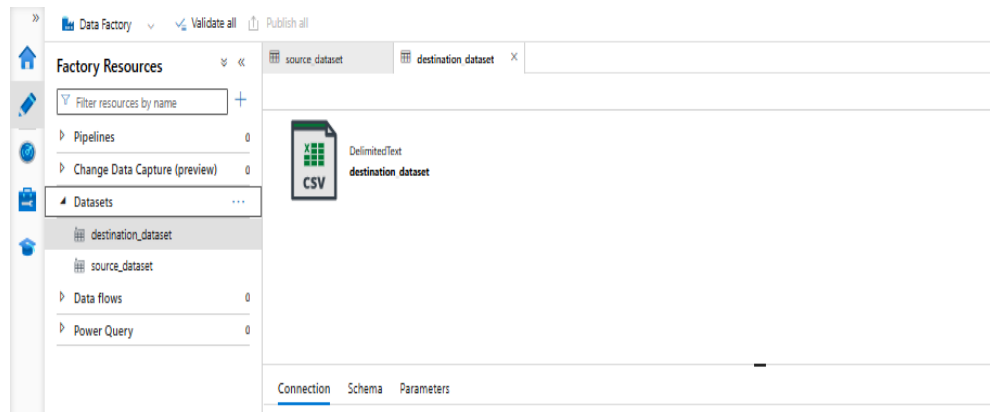
3. Created source_dataset of type DelimitedText (CSV) in ADF Studio under Author → Datasets pointing to source CSV file.

Output-



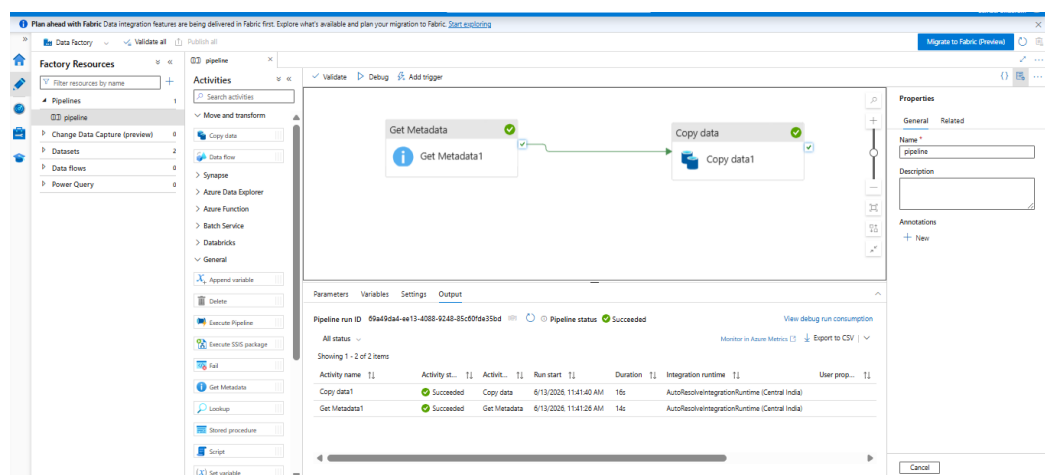
4. Created destination_dataset of type DelimitedText (CSV) in ADF Studio under Author → Datasets pointing to output folder.

Output-



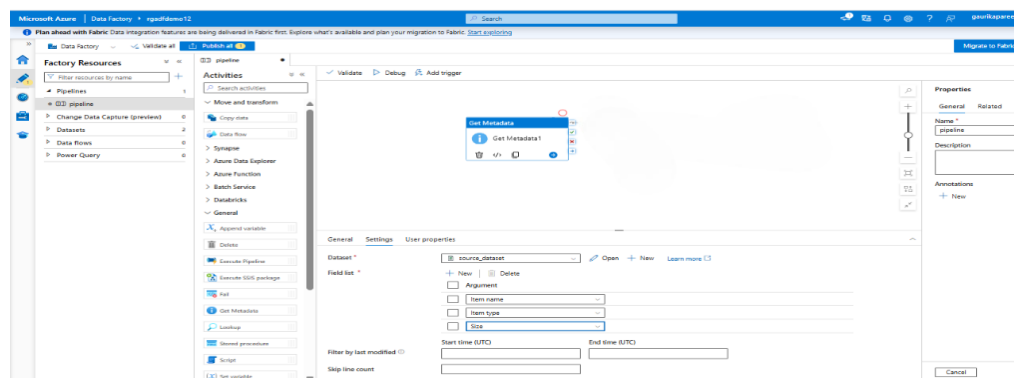
- Pipeline "pipeline" successfully executed with Get Metadata1 → Copy Data1 both showing Succeeded status in Output tab.

Output-



- Configured Get Metadata1 activity with source_dataset and field list (itemname, itemType, Size) for metadata validation.

Output-



- Monitor section showing all pipeline runs with Succeeded status confirming successful pipeline execution.

Output-

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Pipeline runs

Triggered | Debug | Cancel options | Refresh | Edit columns | [View](#) | [Save](#)

Filter by run ID or name | Chennai, Kolkata, Mu... | Last 24 hours | Pipeline name | All | Status | All | Add filter | X

Showing 1-5 items

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Parameters
pipeline	6/13/2026, 11:41:19 AM	6/13/2026, 11:41:57 AM	38s	Succeeded	Manual trigger	69a49dad-e413-4038-9248-85c0f...	
pipeline	6/13/2026, 11:36:32 AM	6/13/2026, 11:36:52 AM	20s	Succeeded	Manual trigger	10074519-4d55-43d6-9910-11306...	
pipeline	6/13/2026, 11:33:22 AM	6/13/2026, 11:33:37 AM	15s	Succeeded	Manual trigger	38bf5884-2bac-470f-a030-9f6ad...	
pipeline	6/13/2026, 11:32:46 AM	6/13/2026, 11:33:08 AM	22s	Succeeded	Manual trigger	759664d2-f521-477b-811f-8d6744...	
pipeline	6/13/2026, 11:29:24 AM	6/13/2026, 11:29:46 AM	22s	Succeeded	Manual trigger	0e93774f-68d5-48d5-ea31-52168...	

Last refreshed 0 minutes ago

8. **Output verification**— Sample-Superstore.csv successfully copied to cont/output/ folder in Blob Storage.

Final Output- In this Mini Project, a complete end-to-end data pipeline was built using Azure Data Factory to read, validate and copy a CSV file from Blob Storage.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > storage12accounttp | Containers

cont ...

Container

Search

+ Add Directory | Upload | Refresh | Delete | Copy | Paste | Rename | Acquire lease | Break lease | Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Leas
cont > output					
Sample - Superstore.csv	6/13/2026, 11:41:52 AM	Hot (Inferred)	Block blob	2.18 MB	Avail

Add or remove resources. See resources of cont in this Blob.